

GIRRAWEE HIGH SCHOOL

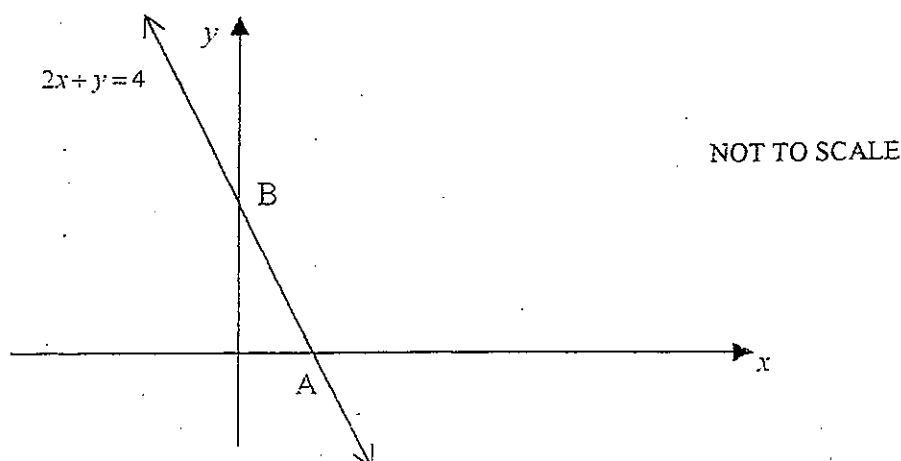
MATHEMATICS

Year 11
Task 4

11th August 2005
Time: 90 minutes

Instructions: Attempt all questions.
Write your answers on your own paper.
All necessary working must be shown.
Marks may be deducted for careless or badly arranged work.
Begin each question on a new page.

Question 1(13 marks)



In the diagram above, the line $2x + y = 4$ cuts the x -axis at A and the y -axis at B.

Copy the diagram into your answer sheet.

- | | |
|---|---|
| a. Find the coordinates of points A and B. | 2 |
| b. Find the perpendicular distance of the point C(5,2) from the line $2x + y = 4$. | 3 |
| c. Show that the gradient of the line AC is $\frac{2}{3}$. | 2 |
| d. Hence or otherwise find the equation of the line AC. | 2 |
| e. Find the distance AB. | 2 |
| f. Find the exact area of $\triangle ABC$. | 2 |

Question 2(12marks)

- a. Find the equation of the line that has an angle of inclination of 45° with the x -axis and a y -intercept of -1 . 2
- b. Prove that the line $4x + 3y - 20 = 0$ is a tangent to the circle $x^2 + y^2 = 16$. 4
- c. Find the equation of the straight line through $(-4, -1)$ that passes through the intersection of the lines $2x + y - 1 = 0$ and $3x + 5y + 16 = 0$. 4
- d. Prove that the points $A(1, 2)$, $B(-1, 6)$ and $C(2, 0)$ are collinear. 2

Question 3(12marks)

- a. Evaluate
- (i) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ (ii) $\lim_{x \rightarrow 0} \frac{x^2 + 7x}{x}$ 4
- (iii) $\lim_{x \rightarrow \infty} \frac{x}{x + 3}$ (iv) $\lim_{x \rightarrow 5} \frac{x^3 - 125}{x - 5}$ 4
- b. If $f(x) = 2x^2 - 5$,
- (i) find $f(x+h) - f(x)$. 2
- (ii) differentiate $2x^2 - 5$ from first principles. 2

Question 4 (18marks)

a. Differentiate

(i) $y = 3x^2 + 5x + 1$

(ii) $y = 4x^5 - 2x^7$ 4

(iii) $y = \frac{7}{x}$

(iv) $y = 3\sqrt{x}$ 4

(v) $y = x^2(x+3)$

(vi) $y = \frac{2x^3 + 5x}{x}$ 4

b. Find $\frac{dy}{dx}$ when $x = -2$ if $y = 7x^2 - 5x + 6$. 3

c. If $f(x) = x^3 + 2x^2 - 4x$, find $f'(-3)$. 3

Question 5 (16marks)

Differentiate

a. $y = (2x^2 - 7)(3x + 1)$

b. $y = \frac{x+5}{3x+2}$ 6

c. $y = (5x^2 - 8)^6$

d. $y = 2x\sqrt{8x-5}$ 7

e. $y = (x+3)^2(2x-5)$ 3

Question 6 (17marks)

a. Find the equation of the **tangent** and **normal** to the curve $y = 2x^4 + 4x$ at the point where $x = 1$. 5

b. Find any x values for which the gradient of the tangent to the curve $y = x^2 - 3x - 1$ is parallel to the line $x - 2y - 1 = 0$. 5

c. Find the equation of the tangent to the curve $y = x^3 - x^2 + 2x + 6$ at the point $P(1,8)$. Find the coordinates of point Q where this tangent meets the y -axis and calculate the exact length of PQ . 7

Question 1 (13 marks)

i) $2x + y = 4$

$x \text{ int} \Rightarrow y = 0$

$2x = 4$

$x = 2 \therefore A(2, 0)$

$y \text{ int} \Rightarrow x = 0$

$y = 4 \therefore B(0, 4)$ (2)

ii) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad C(5, 2)$
 $2x + y - 4 = 0$

$= \frac{|2(5) + 1(2) - 4|}{\sqrt{2^2 + 1^2}}$

$= \frac{8}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{8\sqrt{5}}{5} \text{ units}$ (3)

c) $m_{AC} = \frac{2-0}{5-2} = \frac{2}{3}$

d) $y - y_1 = m(x - x_1) \quad m = \frac{2}{3}$
 $y - 0 = \frac{2}{3}(x - 2)$
 $3y = 2x - 4$
 $2x - 3y - 4 = 0$ (2)

e) $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
 $= \sqrt{(-2)^2 + (4)^2}$
 $= \sqrt{20}$
 $= 2\sqrt{5} \text{ units}$ (2)

f) Area of $\triangle ABC$

$A = \frac{1}{2}bh$

$= \frac{1}{2} \times 2\sqrt{5} \times \frac{8}{\sqrt{5}}$

$= 8 \text{ U}^2$ (2)

Question 2 (12 marks)

a) $m = \tan 45^\circ = 1 \quad b = -1$

Eq: $y = x - 1$ (2)

b) distance of tangent from origin

$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad 4x + 3y - 20 = 0 \quad (0, 0)$

$= \frac{|-20|}{\sqrt{4^2 + 3^2}}$

$= \frac{20}{5} = 4 = \text{radius of circle}$

$\therefore 4x + 3y - 20 = 0$ is a tangent to the circle $x^2 + y^2 = 16$.

$P(-4, -1)$ (4)

c) $(2x + y - 1) + k(3x + 5y + 16) = 0$

$(2(-4) - 1 - 1) + k(3(-4) + 5(-1) + 16) = 0$

$-10 - k = 0$

$k = -10$

$\therefore (2x + y - 1) - 10(3x + 5y + 16) = 0$

$2x + y - 1 - 30x - 50y - 160 = 0$

$-28x - 49y - 161 = 0$

$4x + 7y + 23 = 0$ (4)

d) $m_{AB} = \frac{6-2}{-1-1} = \frac{4}{-2} = -2$

$m_{BC} = \frac{0-6}{2-1} = \frac{-6}{1} = -6$ (2)

$\therefore A, B$ and C are collinear

Question 3 (12 marks)

a) i) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ ii) $\lim_{x \rightarrow 0} \frac{x^2 + 7x}{x}$

$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)} = \lim_{x \rightarrow 0} \frac{x(x+7)}{x}$

$= 6$ (2) $= 7$ (2)

iii) $\lim_{x \rightarrow 0} \frac{x}{x+3}$ iv) $\lim_{x \rightarrow 5} \frac{x^3 - 125}{x - 5}$

$= \lim_{x \rightarrow 0} \frac{1}{1 + 3/x}$ $= \lim_{x \rightarrow 5} \frac{(x-5)(x^2 + 5x + 25)}{x - 5}$

$= 1$ (2) $= 75$ (2)

b) $f(x) = 2x^2 - 5$

i) $f(x+h) - f(x)$

$= 2(x+h)^2 - 5 - (2x^2 - 5)$

$= 2(x^2 + 2xh + h^2) - 5 - 2x^2 + 5$

$= 2x^2 + 4xh + 2h^2 - 5 - 2x^2 + 5$

$= 4xh + 2h^2$ (2)

ii) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2}{h}$

$= \lim_{h \rightarrow 0} \frac{h(4x + 2h)}{h}$

$= 4x$ (2)

Question 4 (18 marks)

a) i) $y = 3x^2 + 5x + 1$

$\frac{dy}{dx} = 6x + 5$ (2)

ii) $y = 4x^5 - 2x^7$

$\frac{dy}{dx} = 20x^4 - 14x^6$ (2)

iii) $y = \frac{7}{x} = 7x^{-1}$

$\frac{dy}{dx} = -7x^{-2}$
 $= \frac{-7}{x^2}$ (2)

iv) $y = 3\sqrt{x} = 3x^{1/2}$

$\frac{dy}{dx} = \frac{3}{2}x^{-1/2}$
 $= \frac{3}{2\sqrt{x}}$ (2)

v) $y = x^2(x+3) = x^3 + 3x^2$

$\frac{dy}{dx} = 3x^2 + 6x$ (2)

vi) $y = \frac{2x^3 + 5x}{x}$

$y = 2x^2 + 5$

$\frac{dy}{dx} = 4x$ (2)

b) $y = 7x^2 - 5x + 6$

$\frac{dy}{dx} = 14x - 5$

When $x = -2$, $\frac{dy}{dx} = 14(-2) - 5$
 $= -33$ (3)

c) $f(x) = x^3 + 2x^2 - 4x$

$f'(x) = 3x^2 + 4x - 4$

$f'(-3) = 3(-3)^2 + 4(-3) - 4$

$= 27 - 12 - 4$

$= 11$ (3)

Question 5 (16 marks)

a) $y = (2x^2 - 7)(3x + 1)$

$$\begin{aligned}\frac{dy}{dx} &= Vu' + UV' \\ &= (3x+1)4x + (2x^2-7)3 \\ &= 12x^2 + 4x + 6x^2 - 21 \\ &= 18x^2 + 4x - 21\end{aligned}\quad (3)$$

b) $y = \frac{x+5}{3x+2}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{Vu' - UV'}{V^2} \\ &= \frac{(3x+2)1 - (x+5)3}{(3x+2)^2} \\ &= \frac{3x+2-3x-15}{(3x+2)^2} \\ &= \frac{-13}{(3x+2)^2}\end{aligned}\quad (3)$$

c) $y = (5x^2 - 8)^6$

$$\begin{aligned}\frac{dy}{dx} &= 6(5x^2 - 8)^5 \cdot 10x \\ &= 60x(5x^2 - 8)^5\end{aligned}\quad (3)$$

d) $y = 2x\sqrt{8x-5}$

$$\begin{aligned}\frac{dy}{dx} &= Vu' + UV' \\ &= (8x-5)^{\frac{1}{2}} \cdot 2 + 2x \left[4(8x-5)^{-\frac{1}{2}} \cdot 8 \right] \\ &= 2\sqrt{8x-5} + \frac{8x}{\sqrt{8x-5}} \\ &= \frac{2(8x-5) + 8x}{\sqrt{8x-5}} \\ &= \frac{16x - 10 + 8x}{\sqrt{8x-5}} \\ &= \frac{24x - 10}{\sqrt{8x-5}}\end{aligned}\quad (4)$$

e) $y = (x+3)^2(2x-5)$

$$\begin{aligned}\frac{dy}{dx} &= Vu' + UV' \\ &= (2x-5)[2(x+3)] + (x+3)^2 \cdot 2 \\ &= 2(x+3)(2x-5+x+3) \\ &= 2(x+3)(3x-2)\end{aligned}\quad (3)$$

Question 6 (17 marks)

a) $y = 2x^4 + 4x$ at $x=1$ pt (1,6)

Eq of tangent

$$\frac{dy}{dx} = 8x^3 + 4$$

$$m = 8(1)^3 + 4 = 12$$

$$y - 6 = 12(x - 1)$$

$$y - 6 = 12x - 12$$

$$12x - y - 6 = 0\quad (3)$$

Equation of normal

$$m = -\frac{1}{12} \text{ pt } (1,6)$$

$$y - 6 = -\frac{1}{12}(x - 1)$$

$$12y - 72 = -x + 1$$

$$x + 12y - 73 = 0\quad (2)$$

b) $y = x^2 - 3x - 1$

$$\frac{dy}{dx} = 2x - 3$$

$$2x - 3 = \frac{1}{2}$$

$$2x = \frac{7}{2}$$

$$x = 1\frac{3}{4}$$

(5)

c) $\frac{dy}{dx} = 3x^2 - 2x + 2$ (1)

at (1,8) $m = 3(1)^2 - 2(1) + 2 = 3$ (1)

Eq: $y - 8 = 3(x - 1)$

$$3x - y + 5 = 0\quad (2)$$

y intercept = 5

$\therefore P(0,5)$ (1)

$$PQ = \sqrt{(8-5)^2 + (0-1)^2}$$

$$= \sqrt{10} \text{ units.}\quad (2) \quad (7)$$